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CSE A-2

Subject: ProgrammingwithJava

ASSIGNMENT No : 8

TITLE : Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

```
class Parent {
    void display () {
        System.out.println("Parent class method");
    }
}

class Child extends Parent {
    void display() {
        super.display();
        System.out.println("Child class method");
    }
}

public class Main {
    public static void main (String[] args) {
        Child c = new Child();
        c.display();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp8> java Main.java
Parent class method
Child class method
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp8> █
```

Exercise

1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

```
class Employee {
    void display() {
        System.out.println(x: "This is employee class");
    }
}

class Manager extends Employee {
    void display() {
        super.display();
        System.out.println(x: "This is manager class");
    }
}

public class Ex1 {
    Run | Debug
    public static void main (String[] args) {
        Manager m = new Manager();
        m.display();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp8\Exercise> java Ex1.java
This is employee class
This is manager class
```

2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

```
class Vehicle {
    String brand;
    String model;

    Vehicle(String b, String m) {
        this.brand = b;
        this.model = m;
    }
}

class Insurance extends Vehicle {
    Insurance(String b, String m) {
        super(b, m);
    }

    void display() {
        System.out.println("Brand : " + brand);
        System.out.println("Model : " + model);
    }
}

public class Ex2 {
    Run | Debug
    public static void main (String[] args) {
        Insurance i = new Insurance(b: "Hyundai", m: "Exter");
        i.display();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp8\Exercise> java Ex2.java
Brand : Hyundai
Model : Exter
```